

Introduction to Microeconomics

Lecture 8—Consumer Theory: The Budget Set

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The budget set and the budget line for the consumer.

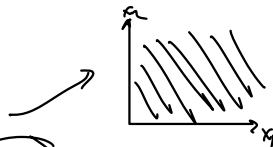
- Goal: represent what is feasible to consume given market prices and income.
- Consumption space: $X = \mathbb{R}_+^n$, n goods, indexed x_i , $i = 1, \dots, n$.
- Prices $p_1, \dots, p_n$ for each good, all non-negative.
- Consumer takes prices as given, i.e., consumer cannot “bargain” for a different price.
- The budget set is the set of all **feasible** consumption bundles given prices and income I of the consumer:

$$B(p_1, \dots, p_n, I) = \{(x_1, \dots, x_n) : \underbrace{p_1 x_1 + p_2 x_2 + \dots + p_n x_n}_{\text{Total Expenditure.}} \leq I\}$$

Expenditure on good 1 (pointing to $p_1 x_1$)
budget (pointing to I)

- Intuitively: Total expenditures must be less than or equal to Income.
- When $p_1 x_1 + \dots + p_n x_n = I$, this corresponds to the **budget hyperplane**, and is the set of consumption bundles that are feasible and exhaust the consumer's income.

The two-good case

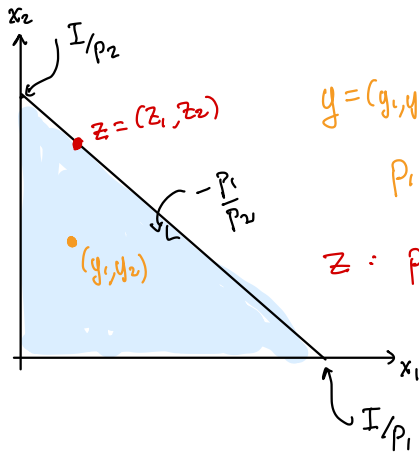


- Suppose we are only considering two goods, i.e., $X = \mathbb{R}_+^2$ goods x_1 and x_2 .
- Prices are p_1 and p_2 and income I .
- The budget set is described by the simple inequality: $p_1x_1 + p_2x_2 \leq I$.
- The budget line is $p_1x_1 + p_2x_2 = I$ is the set of feasible consumption bundles which exhaust the consumer's income.

Graphical Representation

$$p_1 x_1 + p_2 x_2 \leq I \rightarrow \text{BL: } p_1 x_1 + p_2 x_2 = I \quad \rightarrow \left| \text{slope} \right| = \left| -\frac{p_1}{p_2} \right| = \frac{p_1}{p_2} \text{ price ratio}$$

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$



$y = (y_1, y_2)$ is feasible but doesn't exhaust I

$$p_1 y_1 + p_2 y_2 < I$$

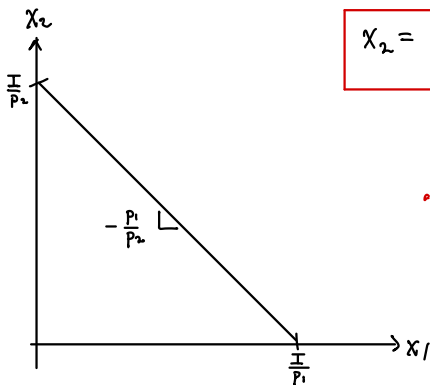
$z : p_1 z_1 + p_2 z_2 = I$ feasible and exhausts I

$\therefore \frac{p_1}{p_2}$ is the $\swarrow$ market O.C. of good 1 (in terms of good 2)

Trade off between the two goods!

$\frac{p_1}{p_2}$: How much it costs, in terms of units of good 2, that I must give up to buy 1 extra unit of good 1

The slope of the budget line



$$X_2 = \frac{I}{P_2} - \frac{P_1}{P_2} X_1$$

$$\rightarrow \text{slope} = -\frac{P_1}{P_2} = \frac{P_1}{P_2} \text{ price ratio}$$

$\therefore \frac{P_1}{P_2}$ is the $\swarrow$ market O.C. of good 1 (in terms of good 2)

Similarly

$\frac{P_2}{P_1}$ is the market O.C. of good 2 (in terms of good 1)

Trade off between the two goods!

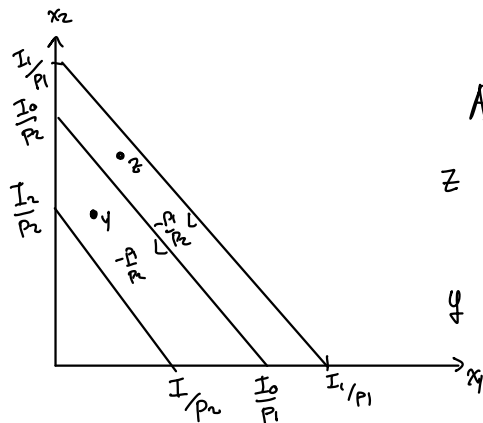
$\frac{P_1}{P_2}$: How much it costs, in terms of units of good 2, that I must give up to buy 1 extra unit of good 1

Changes in Income

$$I_0: p_1 x_1 + p_2 x_2 = I_0$$

say that $I_1 > I_0 > I_2$

$$I_1 = p_1 x_1 + p_2 x_2$$



An ΔI shifts in a parallel way the BL

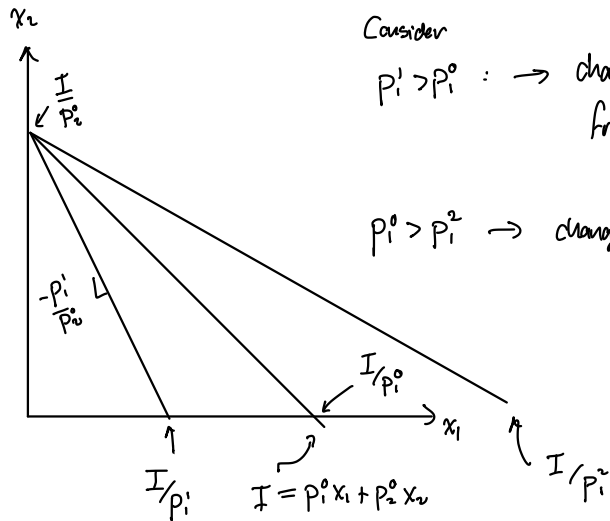
z is not feasible with incomes I_0, I_2

feasible with I_1

y feasible with I_1, I_0

not feasible with I_2

$$P_1^0 X_1 + P_2^0 X_2 = I \quad (\text{original})$$



Changes in p_2

x_2

↑

price decrease in p_2

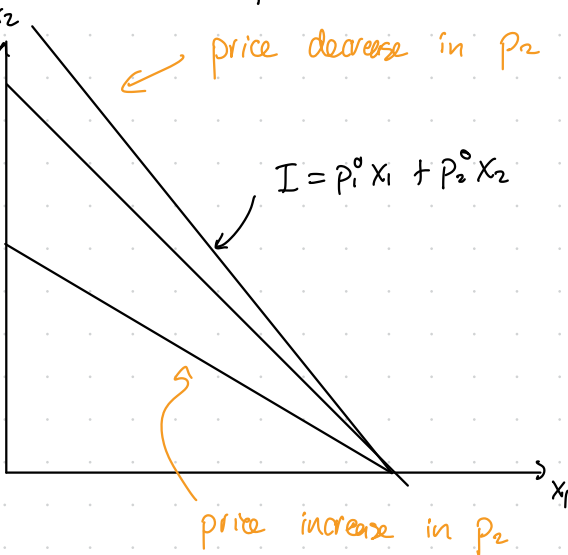
$$I = p_1^0 x_1 + p_2^0 x_2$$

↑

price increase in p_2

→

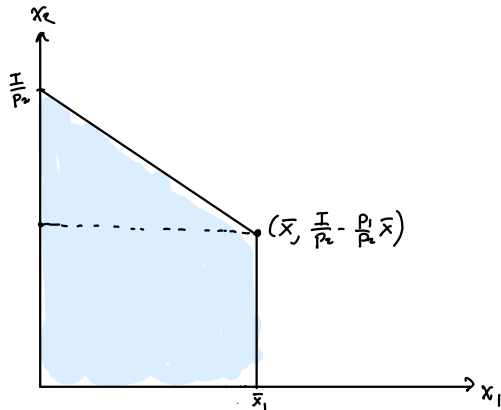
x_1



Non-linear budget constraints

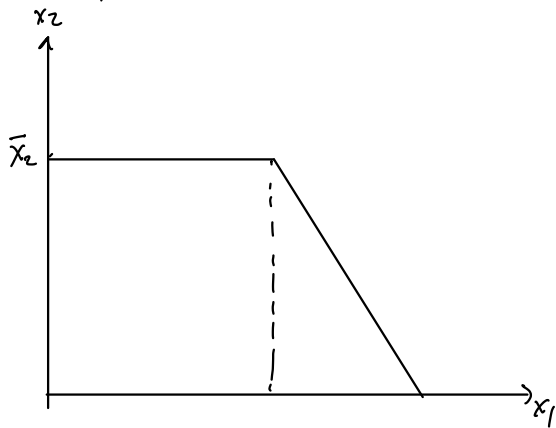
Quantity Restriction : you are not allowed to buy more than $\bar{x}_1$ units of good 1 prices of good 1 & 2 p_1, p_2

$$x_2 = \begin{cases} \frac{I}{p_2} - \frac{p_1}{p_2} x_1 & \text{if } x_1 < \bar{x}_1 \\ [0, \frac{I}{p_2} - \frac{p_1}{p_2} \bar{x}_1] & \text{if } x_1 = \bar{x}_1 \end{cases}$$



Non-linear budget constraints

Quantity restriction for x_2 , $\bar{x}_2$



HW: write here the analytical description

Non-linear budget constraints

Example : Food Stamps Suppose

$$I = \$100, \quad p_F = \$1$$

(price of "other" goods) $p_G = \$1$

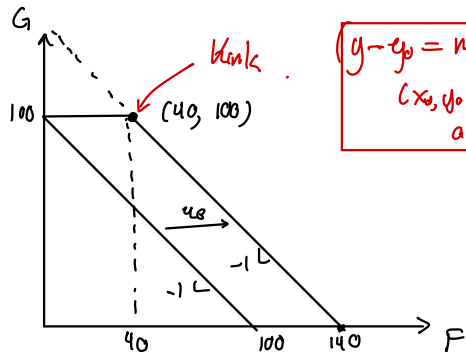
We keep fixed the price of other things G , as $p_G = 1$ is called "normalizing", this good is called the numeraire.

Budget Line : $100 = F + G$

- You receive \$40 worth of Food stamps.

Eq. BL with Food stamps

$$G = \begin{cases} 100 & 0 \leq F \leq 40 \\ 140 - F & 40 < F \leq 140 \end{cases}$$



$$y - y_0 = m(x - x_0)$$

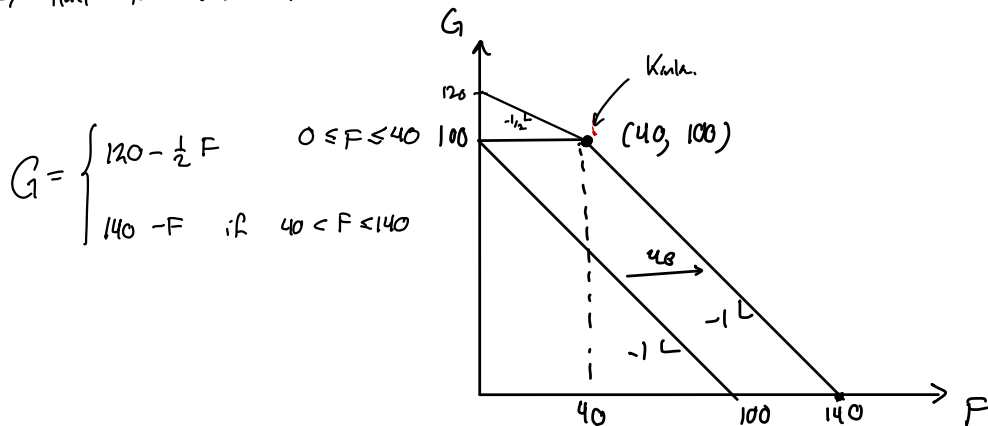
(x_0, y_0) is a point

Non-linear budget constraints

What if there is a Black Market for Food Stamps?

- You can sell Food Stamps for \$0.50 each?

↳ Half the value in terms of food.



Non-linear budget constraints

Quantity Based Discount.

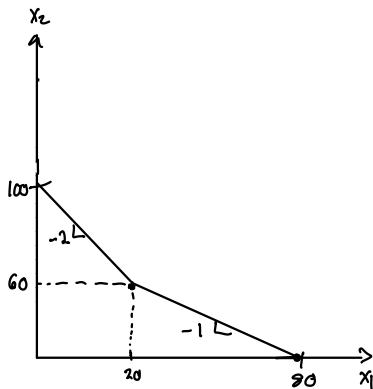
→ Quantity discount for large buyers.

Suppose $p_2 = 1$, $I = 100$

$p_1 = 2$ for $0 \leq x_1 \leq 20 \Rightarrow \frac{p_1}{p_2} = 2$

$p_1 = 1$ for $x_1 > 20 \Rightarrow \frac{p_1}{p_2} = 1$

$$x_2 = \begin{cases} 100 - 2x_1 & \text{if } 0 \leq x_1 \leq 20 \\ 80 - x_1 & \text{if } 20 < x_1 \leq 80 \end{cases}$$



$$(20, 60) \rightarrow \tilde{I} = c_1(20) + c_2(60) \rightarrow x_1 + x_2 = 80 \\ = 80$$

Where are we going: Consumer Problem

$$\max_{(x_1, \dots, x_n)} U(x_1, \dots, x_n)$$

$$\text{subject to: } p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq I$$

Parameters

$$p_1, \dots, p_n$$

$$I$$

Choice variables

$$x_1, \dots, x_n$$